

Question 1 :

$$a_k = \frac{1}{2} \int_{-2/2}^{2/2} x(t) \cos\left(\frac{k 2\pi t}{2}\right) dt \quad , \text{Here } b_k = 0, \text{ because } x(t) = x(-t) \text{ is a Even signal}$$

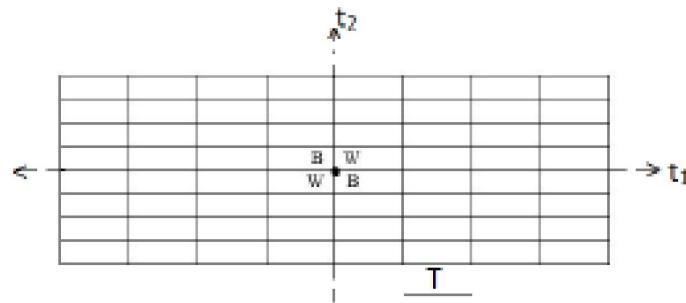
$$\text{Also } a_0 = \frac{1}{2} \int_{-2/2}^{2/2} x(t) dt = 0$$

So,

$$a_k = \frac{1}{2} \left\{ \int_{-1}^{-1/2} x(t) \cos\left(\frac{k 2\pi t}{2}\right) dt + \int_{-1/2}^{1/2} x(t) \cos\left(\frac{k 2\pi t}{2}\right) dt + \int_{1/2}^1 x(t) \cos\left(\frac{k 2\pi t}{2}\right) dt \right\}$$

$$a_k = \frac{1}{2} [2 \sin\left(\frac{k\pi}{2}\right) - \sin(k\pi)]$$

Question 2:



We assume that the chess board is of infinite size, so that we can have Fourier series.

$$x(t_1, t_2) = \sum_n \sum_m a_{mn} e^{jmw_x t_1} e^{jnw_y t_2}$$

$$a_{mn} = \left(\frac{1}{T_1 T_2}\right) \int_{T_1} \int_{T_2} x(t_1, t_2) e^{-jmw_x t_1} e^{-jnw_y t_2} dt_1 dt_2$$

$$T_1 = 2, \quad T_2 = 2, \quad w_x = \pi, \quad w_y = \pi$$

$$x(t_1, t_2) = u(t_1)u(-t_2) + u(-t_1)u(t_2)$$

$$a_{mn} = \left(\frac{2}{mn\pi^2}\right), \quad \text{If } m \text{ \& } n \text{ are both odd.}$$

$$= 0, \quad \text{Otherwise}$$

$$x(t_1, t_2) = \sum_n \sum_m \frac{2}{mn\pi^2} e^{-jm\pi t_1} e^{jn\pi t_2} \quad , \text{For } m \text{ and } n \text{ odd}$$

$$= 0 \quad , \text{otherwise}$$

Question 3: Fourier coefficients of signal $g(t)$

$$\begin{aligned} g_k &= (1/T) \int_T g(t) e^{-jk(2\pi/T)t} dt \\ &= (1/4) \int_{-2}^2 g(t) e^{-jk(\pi/2)t} dt \quad [\text{fundamental period } T=4] \end{aligned}$$

when $k = 0$,

$$g_k = (1/4) \int_{-2}^2 g(t) dt = 0$$

when $k \neq 0$,

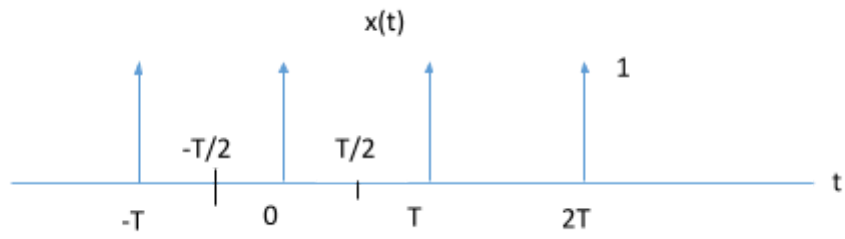
$$\begin{aligned} g_k &= (1/4) \left[\int_{-2}^0 g(t) e^{-jk(\pi/2)t} dt + \int_0^2 g(t) e^{-jk(\pi/2)t} dt \right] \\ &= (1/4) \left[(-1/2) \int_{-2}^0 e^{-jk(\pi/2)t} dt + (1/2) \int_0^2 e^{-jk(\pi/2)t} dt \right] \\ &= (1/8) \left[\int_0^2 e^{-jk(\pi/2)t} dt - \int_{-2}^0 e^{-jk(\pi/2)t} dt \right] \\ &= (1/8) \left[\frac{1}{-jk(\pi/2)} (e^{-jk\pi} - 1 - 1 + e^{-jk\pi}) \right] \\ &= [1/2jk\pi] [1 - e^{-jk\pi}] \\ &= [1/k\pi] \left[\frac{e^{jk(\pi/2)} - e^{-jk(\pi/2)}}{2j} \right] e^{-jk(\pi/2)} \\ &= \frac{\sin(k\pi/2)}{k\pi} e^{-jk(\pi/2)} \end{aligned}$$

As $\frac{dx(t)}{dt} = g(t)$, from the differentiation property of Fourier series,

$$\begin{aligned} jk\omega x_k &= g_k \\ x_k &= \frac{\sin(k\pi/2)}{(jk\pi)k\pi} \cdot e^{-jk(\pi/2)} = \frac{2\sin(k\pi/2)}{jk^2\pi^2} \cdot e^{-jk(\pi/2)} \quad \text{for } k \neq 0 \\ &= 0 \quad \text{for } k = 0 \end{aligned}$$

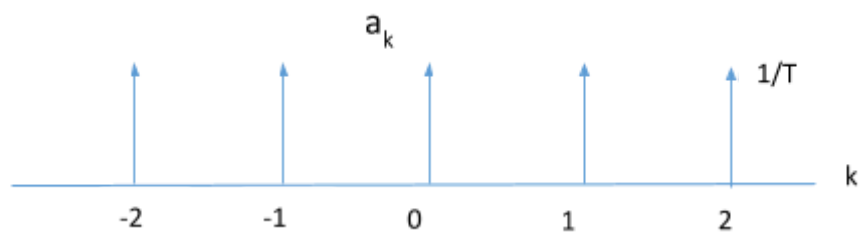
Question 4:

a.
$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

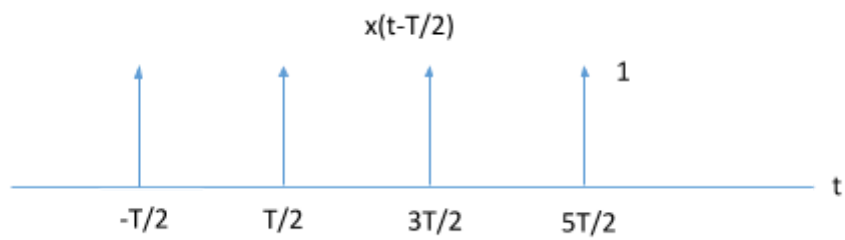
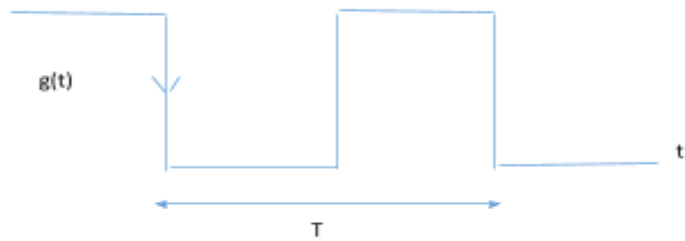


$$x_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \delta(t) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{T}$$



b.



$$\begin{aligned}\frac{dg(t)}{dt} &= \sum_{k=-\infty}^{\infty} \delta(t-kT) - \sum_{m=-\infty}^{\infty} \delta\left(t - \left(m + \frac{1}{2}\right)T\right) \\ &= x(t) - x(t - (T/2))\end{aligned}$$

Hence, using the differentiation and time shifting properties of Fourier series coefficients with $g(t)$ and $x(t)$ having fundamental period $T = 4$.

$$\begin{aligned}jk\omega_0 g_k &= x_k - x_k e^{-jk\omega_0(T/2)} \\ &= x_k (1 - e^{-jk\pi}) \\ g_k &= \frac{x_k}{jk(2\pi/T)} (1 - e^{-jk\pi}) \\ g_k &= \frac{(1/T)}{jk(2\pi/T)} (1 - e^{-jk\pi}) \\ g_k &= \frac{1}{jk(2\pi)} e^{-jk\pi/2} (e^{jk\pi/2} - e^{-jk\pi/2}) \\ g_k &= [1/k\pi] \left[\frac{e^{jk(\pi/2)} - e^{-jk(\pi/2)}}{2j} \right] e^{-jk(\pi/2)} \\ g_k &= \frac{\sin(k\pi/2)}{k\pi} e^{-jk(\pi/2)}\end{aligned}$$

Question 5:

Fundamental frequencies will be the same, $\omega_1 = \omega_2$.

$$\begin{aligned}x_2(t) &= x_1(1-t) + x_1(t-1) \\ x_2(t) &= x_1(-t+1) + x_1(t-1) \\ x_1(-t+1) &\leftrightarrow a_{-k} e^{-jk\omega_1} \\ x_1(t-1) &\leftrightarrow a_k e^{-jk\omega_1} \\ b_k &= [a_k + a_{-k}] e^{-jk\omega_1}\end{aligned}$$

Question 6:

a) $x(t)$ is real and odd

hence, a_k are imaginary and odd and $a_0 = 0$

b) period is $T=2$

$$\omega_0 = \frac{2\pi}{2} = \pi$$

$$c) \quad x(t) = a_1 e^{j\omega_1 t} + a_{-1} e^{-j\omega_1 t}$$

$$|a_1|^2 + |a_{-1}|^2 = 1$$

$$|a_1|^2 = \frac{1}{2}$$

$$a_1 = \frac{1j}{\sqrt{2}}, \frac{1(-j)}{\sqrt{2}}$$

$$a_1 = \frac{j}{\sqrt{2}}, \frac{-j}{\sqrt{2}}$$

$$x(t) = \pm \left[\frac{j}{\sqrt{2}} e^{j\pi t} + \frac{(-j)}{\sqrt{2}} e^{-j\pi t} \right]$$

$$x(t) = \pm \frac{j}{\sqrt{2}} 2j [e^{j\pi t} - e^{-j\pi t}] / 2j$$

$$x(t) = \pm \sqrt{2} \sin(\pi t)$$

Question 7:

$$\frac{2\pi}{\pi} * k = 250$$

$$k = 17.85$$

$$a_k = 0 \text{ for } |k| < 18$$

Question 8:

$$(a) \quad x(t-t_0) + x(t+t_0)$$

$$\text{Fourier Series:} \quad x(t-t_0) + x(t+t_0) \longleftrightarrow a_k e^{-jk\omega_0 t_0} + a_k e^{jk\omega_0 t_0}$$

$$(b) \quad \text{Ev}\{x(t)\} = [x(t) + x(-t)]/2$$

Fourier Series representation:

As $FS\{x(t)\} \rightarrow a_k$ hence we have the fourier series representation of $x(-t)$ is equal to a_{-k}

as we have $FS(x(-t)) = 2/T * \int_0^T a'_k e^{-2\pi kt/T}$ where a'_k is the fourier series coefficient of $x(-t)$

if we choose a'_k as a_{-k} by putting k as $-k$ in the above equation we find the familiar expression of Fourier Series coefficients. Hence

$$F.S\{\text{Ev}\{x(t)\}\} \rightarrow (a_k + a_{-k})/2$$

$$(c) \quad \text{Re}\{x(t)\}$$

Now, $\text{Re}\{x(t)\} = [x(t) + \overline{x(t)}]/2$, Here $\overline{x(t)}$ represents the conjugate of $x(t)$

So for the Fourier Series representation:

As $FS\{x(t)\} \rightarrow a_k$ so the fourier series representation of $\overline{x(t)}$ is equal to $\overline{a_{-k}}$

as they are given by $a'_k = 2/T * \int_0^T \overline{x(t)} e^{i2\pi nt/T}$ where a'_k is the fourier series coefficient of $\overline{x(t)}$

and taking conjugate both sides we get that $\overline{a'_k} = 2/T * \int_0^T x(t) e^{-i2\pi nt/T}$ after which we just need

to replace t by -t as a variable to get the fourier series coefficients in terms of a_k

So we get $F.S. \{Re(x(t))\} \rightarrow (a_k + \overline{a_{-k}})/2$

$$(d) \frac{d^2 x}{dt^2} \rightarrow (jkw_0)^2 a_k$$

$$(e) x(3t-1) \rightarrow a_k e^{-jk w_0}$$

The period of the given signal is 1/3rd of the period of the signal x(t)

Question 9

$$(a) a_k = a_{k+2}$$

$$a_k = \frac{1}{T} \int x(t) e^{-jktw_0} dt$$

$$\text{While we also have } a_{k+2} = \frac{1}{T} \int x(t) e^{-jk(t+2)w_0} dt$$

$$\text{Which gives us } x(t) = x(t) e^{-j4\pi t/3}$$

$$\text{Hence } x(t) \text{ can be non-zero only when, } e^{-4j\pi t/3} = 1$$

This implies that;

$$e^{-j4\pi t/3} = e^{j2\pi m}$$

$$\text{Hence } \frac{4\pi t}{3} = 2\pi m$$

$$\text{and, } t = \frac{3}{2} m$$

$$t = 0, \pm 1.5, \pm 3, \dots$$

$x(t) = a\delta(t) + b\delta(t-1.5) + c\delta(t+1.5)$ (Period is 3 we do not need to consider the signal beyond the range of [-1.5, 1.5])

(b) As we have from $a_k = a_{-k}$ from which we get $b = c$ (can use a method similar to Q8 (a) to get the relation between b and c and a point to note here is the period of the signal which is 3)

$$x(t) = a\delta(t) + b\delta(t+1.5) + b\delta(t-1.5)$$

$$(c) \text{ Since the } \int_{-0.5}^{0.5} a\delta(t) dt = 1 \Rightarrow a = 1$$

(d) Since the $\int_1^2 b\delta(t)dt = 2 \Rightarrow b = 2$

Hence, the given signal $x(t)$ is: $x(t) = \delta(t) + 2\delta(t-1.5) + 2\delta(t+1.5)$

Question 10:

[NOTE: The part (a) in problem sheet represents the question itself]

(b) $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$ $\omega_0 = \frac{2\pi}{T}$. The constraint is that for all even k , $a_k = 0$

$x(t + \frac{T}{2}) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0(t + \frac{T}{2})}$. For odd values of k , $e^{jk\omega_0(\frac{T}{2})} = e^{jk\pi} = -1$

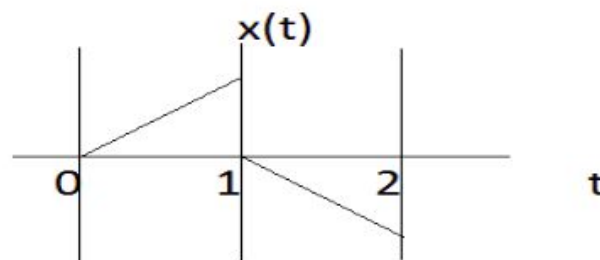
Therefore, $x(t + \frac{T}{2}) = -x(t)$

(c) $a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$

Put $t = p + \frac{T}{2}$. Then, $a_k = \frac{(-1)^{k+1}}{T_0} \int_{T_0} x(p) e^{-jk\omega_0 p} dp \Rightarrow a_k = (-1)^{k+1} a_k$

Therefore, for even values of k , $a_k = 0$

(d)



Here $T = 2$. Thus, $\frac{T}{2} = 1$. Thus, $x(t) = -x(t+1)$. Using this, we obtain the diagram shown above

(e) The fundamental period would be $T/2$.

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$\Rightarrow x(t + \frac{T}{2}) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0(t + \frac{T}{2})}$$

$$\Rightarrow x(t + \frac{T}{2}) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{jk\pi} \quad \text{Since, only even values of 'k' are non-zero, } e^{jk\pi} = 1$$

$$\text{Thus, } x(t + \frac{T}{2}) = x(t)$$

(f) $x(t) = a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t} + \dots$. Clearly the fundamental period is T .

For $x(t) = a_1 e^{jk\omega_0 t} + a_{-1} e^{-j\omega_0 t} + \dots$, then too the fundamental period would be T . If both these conditions are absent, then for any two coefficients, a_m and a_n $m = pn$, where m , n and p are integers. Then the fundamental period would be $\frac{T}{n}$.

Question 11:

The unknown FS coefficients are a_{-1} , a_1 , a_{-2} and a_2

Since $x(t)$ is real $a_1 = a_{-1}^*$ and $a_2 = a_{-2}^*$

Since a_1 is real $a_1 = a_{-1}$

$$x(t) = A_1 \cos(\omega_0 t) + A_2 \cos(2\omega_0 t + \theta) \quad \text{where } \omega_0 = 2\pi/6$$

From this we get

$$x(t - 3) = A_1 \cos(\omega_0 t - 3\omega_0) + A_2 \cos(2\omega_0 t + \theta - 6\omega_0)$$

Now if we need $x(t) = -x(t - 3)$ then $3\omega_0$ and $6\omega_0$ should both be odd multiples of π .

Therefore $a_2 = a_{-2} = 0$ and $x(t) = A_1 \cos(\omega_0 t)$

Using Parseval's relation

$$\sum_{k=-\infty}^{\infty} |a_k|^2 = |a_1|^2 + |a_{-1}|^2 = 1/2$$

[NOTE: In problem sheet 11(e), there is a typo and the correct equation is.

$$(1/6) \int_{-3}^3 |x(t)|^2 dt = 1/2]$$

Then $|a_1| = 1/2$ since a_1 is positive

$$a_1 = a_{-1} = 1/2$$

$$x(t) = \cos(\pi t/3)$$

Question 12:

- a) Pair (a) and (b) are orthogonal, Pair (c) and (d) are not orthogonal.
- b) Orthogonal but not orthonormal $A_m = 1/w_0$
- c) Orthonormal
- d) We have

$$\int_{t_0}^{t_0+T} e^{jm w_0 t} e^{-jn w_0 t} dt = \frac{e^{j(m-n)w_0 t_0} (e^{j(m-n)2\pi} - 1)}{(m-n)w_0}$$

$$= 0, \text{ when } m \neq n$$

$$= jT, \text{ when } m = n$$

Function are orthogonal but not orthonormal

e) We have

$$\int_{-T}^T x_e(t) x_o(t) dt = 1/4 \int_{-T}^T [x(t) + x(-t)][x(t) - x(-t)] dt$$

$$= 1/4 \int_{-T}^T x^2(t) dt - 1/4 \int_{-T}^T x^2(-t) dt$$

$$= 0$$

$$f) \int_a^b \frac{1}{\sqrt{a_k}} \frac{\Phi_k(t)}{\sqrt{a_l}} \Phi_l(t) dt = \frac{1}{\sqrt{a_k a_l}} \int_a^b \Phi_k(t) \Phi_k^*(t) dt$$

$$= 0, k \neq l$$

$$1, k = l$$

Therefore, functions are orthonormal